ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO ROBOTICS

PAPER 2 · PRACTICE C

MARKS: 80 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET second paper (practical, 80 marks): design a breadboard circuit and flowchart for an automatic greenhouse window, write its Arduino program, then change a given pedestrian-crossing program.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of THREE questions. Answer ALL the questions.
2. This is a practical paper. Use Fritzing (or another circuit program) for circuits, draw.io (or similar) for flowcharts and the Arduino IDE for programs. If you do not have a computer, draw the circuits and flowcharts neatly by hand and write the programs in your answer book.
3. Save your work regularly. Name every file with the question number and your name or examination number.
4. No circuits need to be built physically; designs are done on the computer or on paper.
5. Programs are not marked strictly on syntax: marks are given for correct logic, correct use of pins and functions, and comments.
6. Arduino Uno pins: digital pins 0–13 (PWM on 3, 5, 6, 9, 10 and 11), analogue inputs A0–A5, 5 V, 3,3 V and GND.

QUESTION 1

> A small greenhouse gets too hot on sunny days. The owner wants a prototype that opens a roof window automatically. A temperature sensor measures the air temperature, a servo motor opens the window, and a red LED shows when the window is open. A push button lets the owner force the window closed (for example when it rains).

The following components are available:

- Arduino Uno
- TMP36 temperature sensor (output 0,5 V at 0 °C, rising 10 mV for every °C)
- SG90 servo motor (signal, 5 V and GND wires)
- Red LED and a 220 Ω resistor
- Push button and a 10 kΩ resistor
- Breadboard and jumper wires
- USB cable or 9 V battery for the Arduino

1.1 Using the given components, design the breadboard circuit for the greenhouse window. (14)
HINT: The circuit must include the correct wiring of the TMP36 to an analogue input; the servo signal wire to a PWM pin; the LED with its series resistor; the push button with a pull-down resistor; and correct 5 V and GND connections for every component.

1.2 Draw a flowchart that shows the logic of the greenhouse window. HINT: The flowchart must include starting up and setting the pins; a loop that keeps reading the temperature; decisions that open the window fully above 30 °C, half-open it from 25 °C to 30 °C and close it below 25 °C; the override button that always closes the window; and a check for an impossible sensor reading. (14)

1.3 Save and print (or neatly present) the breadboard circuit and the flowchart. (2)

[30]

QUESTION 2

You must now write the program for the greenhouse window in QUESTION 1. The TMP36 is connected to A0, the servo to pin 9, the red LED to pin 7 and the override button to pin 2 (HIGH when pressed).

2.1 Write an Arduino program that does the following: (18)

• Sets up the servo, LED and button pins, and starts the serial monitor.

• Reads the TMP36 continuously and converts the reading to °C, using: voltage = reading × 5,0 ÷ 1 023 and temperature = (voltage − 0,5) × 100.

• Opens the window fully (servo at 90°) above 30 °C, half-opens it (45°) from 25 °C to 30 °C, and closes it (0°) below 25 °C.

• Switches the red LED on whenever the window is open (more than 0°).

• Always closes the window and switches the LED off while the override button is pressed.

• Prints "Sensor error" and closes the window if the temperature is below −40 °C or above 125 °C.

• Is well commented, so the logic is clear.

2.2 Save and print (or neatly present) the program. (2)

[20]

QUESTION 3

The program below controls a pedestrian crossing on a robot test track. When the button is pressed, the cars' light changes from green to yellow to red, and the WALK light comes on so people can cross.

```
// Pedestrian crossing on a robot test track
const int carRed = 2;
const int carYellow = 3;
const int carGreen = 4;
const int walkLed = 5;
const int button = 7;

void setup() {
  pinMode(carRed, OUTPUT);
  pinMode(carYellow, OUTPUT);
  pinMode(carGreen, OUTPUT);
  pinMode(walkLed, OUTPUT);
  pinMode(button, INPUT_PULLUP);   // internal pull-up: LOW when pressed
  digitalWrite(carGreen, HIGH);    // cars may go
}

void loop() {
  if (digitalRead(button) == LOW) {   // button pressed
    crossing();
  }
}

void crossing() {
```

```
  digitalWrite(carGreen, LOW);
  digitalWrite(carYellow, HIGH);
  delay(2000);
  digitalWrite(carYellow, LOW);
  digitalWrite(carRed, HIGH);
  digitalWrite(walkLed, HIGH);      // people may cross
  delay(5000);
  digitalWrite(walkLed, LOW);
  digitalWrite(carRed, LOW);
  digitalWrite(carGreen, HIGH);     // cars may go again
}
```

3.1 Change the program so that, after the WALK light has been on steadily for 5 seconds, it flashes on and off (every half second) for a further 3 seconds before the cars get green. (9)

3.2 Add a buzzer on pin 8 that beeps while the WALK light is on, so that people who cannot see well know when to cross. (9)

3.3 Change the program so that, after a crossing, the button is ignored for 10 seconds, giving the cars green for at least that long. (9)

3.4 Save and print (or neatly present) your final program. (3)

[30]

TOTAL: 80

END OF QUESTION PAPER